Some geometric and topological questions for hybrid systems arising in SLAM with coarse measurements

Daniel Liberzon
UIUC

Abstract

In this talk we discuss hybrid control tasks such as state estimation, localization and mapping in an unknown environment with coarse and discontinuous measurements, and their underlying information requirements. We employ the observability decomposition for linear systems, as well as its differential-geometric counterpart for nonlinear systems, to gain a better understanding of possibilities and limitations inherent to these tasks. For the case of multiple landmarks, this leads us to the problem of decomposing a piecewise constant function into a sum of indicator functions, whose number and location are not a priori known. We investigate topological properties of such decompositions, with a particular focus on the ones possessing robustness to perturbations and with the goal of enumerating and constructing all such decompositions.